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Improved Field Production Performance Through Efficient Real-Time Decisions Using Advanced Reservoir Mapping Solutions, Case Studies in Offshore Abu Dhabi

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Abstract

This study highlights significant production gains achieved through the application of an advanced ultra-deep electromagnetic (UDAR) logging tool in mature offshore carbonate reservoirs. By addressing challenges related to fluid distribution and horizontal well placement, the tool enhanced Geosteering precision and enabled more informed decisions about well trajectory, avoiding unwanted fluid contacts. This precision was crucial in optimizing the design of the lower completion, including decisions regarding blank pipe lengths and calling an early Total Depth (TD) after mapping the conductive zones and geological anomalies which resulted in well life extension. The utilization of UDAR tool in these case studies led to substantial value creation in production, amounting to \$294 million in 2024. This study focuses on three cases and discusses how the tool's ability to reduce geological uncertainty and mapping reservoir zones was instrumental in achieving these production gains, which are critical for future field development in such mature carbonate settings. The primary challenge was to maintain the well trajectory of the three wells within the optimum zone while accurately identifying conductive zones. This precision was essential to blank out unnecessary intervals in the oil producer wells (A, B, and C) and optimize the fit to purpose completion design to preserve the well life. Additionally, post-rig operations conducted by barge were planned to verify the effectiveness of the completion deployment, ensuring it met our operational requirements and aligned with our objectives. During these post-rig operations, assessing the production performance was crucial to demonstrate the effectiveness of the UDAR tool utilization.

In each case, the major challenges highlighted by Reservoir Engineers prior drilling were successfully addressed. In Well A, an early TD was called after the UDAR mapped the increase of a low resistivity zone from the bottom of the target, delaying water influx and increasing well life. In Well B, the UDAR mapped the low resistivity zone below the target, leading to a move to the secondary target zone and blanking that interval in the completions stage. In Well C, the well's trajectory was placed in the optimum zone, minimizing risks related to formation dipping and conductive layers above the trajectory. Post-barge operations indicated dry oil in all three wells, delivering expected rates and extending well life in the absence of artificial lift. In contrast, nearby wells are either suffering or unable to flow due to high water

cut. For effective reservoir management during complex oil wells drilling in water risky areas, a proactive geosteering approach using UDAR inversions has shown significant benefits over traditional strategies that rely on shallow directional measurements. This method improves reservoir access, accelerates drilling operations, and streamlines the well completion process, reducing overall costs and operational complexity.

Introduction

Mature carbonate reservoirs in the Middle East pose numerous challenges for horizontal well placement, including low resistivity contrast, heterogeneous porosity, and unpredictable water encroachment. These reservoirs often exhibit complex diagenetic overprints, tight streaks interbedded with vugular Porosity and poor correlation between offset well log responses. Conventional geosteering techniques using shallow resistivity and gamma ray alone frequently leading to sub-optimal placement or early water breakthrough.

Historically, pilot holes and conventional LWD tools were used to mitigate these uncertainties, but their limited depth of investigation often provided only partial insights into formation heterogeneities. Moreover, reactive Geosteering decisions based solely on near-bit data can be delayed, especially in structurally faulted or rapidly changing environments. As operators look to prolong field life, maximize remaining reserves and reduce drilling costs, there is a critical need for technology that enables proactive decision-making.

In this study, we present three case examples from the Offshore field where Reservoir Management decisions and UDAR played a pivotal role in enhancing long-term production. In each of the cases below, UDAR successfully identified fluid boundaries and potential water encroachment zones in real time. This early detection allowed for timely adjustments in well placement and completion strategies, ultimately safeguarding oil productivity and extending the economic life of the wells. These results underscore the value of UDAR as a strategic asset in the development and management of mature reservoirs.

Technology Overview

The introduction of Ultra-Deep Azimuthal Resistivity (UDAR) tools addresses this challenge by providing a broader depth of investigation and 3D visualization capabilities of subsurface resistivity profiles. These tools provide an opportunity to 'look around' the bit, enabling a transformational shift in how horizontal wells are drilled, steered, and completed in carbonate reservoirs.

UDAR tools operate at lower frequencies (2-64 kHz) with long transmitter-receiver spacing (up to 135 ft) enabling depth of investigation up to 100 ft from the borehole ([Khemissa et al., 2020](#)). Multi-layer inversion algorithms provide azimuthally sensitive 1D and 2D resistivity images that help visualize both structural and fluid boundaries in real-time.

The combination of UDAR with conventional LWD tools such as azimuthal gamma ray, density-neutron, and shallow resistivity enhances geosteering workflows and supports more accurate placement in reservoir sweet spots ([Al-Dhaferi et al., 2014](#); [Goraya et al., 2024](#)).

Geological Context and Operational Challenges

The reservoirs encountered in this study are part of the Lower Cretaceous carbonate platform, typically deposited in a shallow marine environment characterized by tidal flats, shoals and deeper lagoonal settings. The depositional architecture results in pronounced lateral and vertical heterogeneities in porosity and permeability.

Waterflooding has been the principal recovery mechanism in these mature fields for over three decades. However, water slumping, capillary pressure imbalances, and thief zones have led to uneven sweep efficiency. Past field data show extensive water holdup in high-permeability zones and reverse coning due to drawdown gradients ([Cunningham and Chaliha, 2002](#)).

Conventional geosteering in these environments is impeded by the low confidence markers or insufficient vertical resistivity contrast. Furthermore, the decision to call TD or blank out low resistivity zones is

often made with limited confidence. Consequently, some wells suffer from early water cut, uneven sweep conformance or completion issues.

Case Study - Well A

Pre-Well Modeling

The objective was to drill the 6-inch lateral section within Target A, a carbonate reservoir, while utilizing Ultra-Deep Azimuthal Resistivity (UDAR) to map potential water approaching to the wellbore. This would also support real-time geosteering to ensure optimal wellbore placement within the most productive zone, away from low-resistivity waterbearing intervals.

During the pre-well planning phase, nearby offset well 1-01D was selected as the primary reference due to its proximity and geological similarity to the target well. This well was considered the most representative for modeling purposes. Analysis of its resistivity profile revealed no indication of low-resistivity zones either above or below the target interval, Zone A. However, due to the potential uncertainties, a flooded scenario was incorporated into the inversion modeling.

Pre-well modeling confirmed the feasibility of detecting low-resistivity zones beneath the wellbore along the planned trajectory. Extensive inversion modelling was conducted to determine the most effective Bottom Hole Assembly (BHA) configuration, including optimal transmitter-receiver spacing and operating frequency. The modeling results indicated that the ideal setup consisted of two receivers: RX1 positioned 25 ft from the transmitter (TX), and RX2 placed 75 ft from the transmitter.

Following these results, the BHA for the 6-inch section included Logging While Drilling (LWD) tools such as UDAR, propagation resistivity and neutron porosity sensors. The propagation resistivity tool was specifically positioned between RX1 and RX2 to enhance signal resolution and fluid boundary detection. Additionally, gamma ray sensors and density were placed close to the bit for near field reservoir mapping.

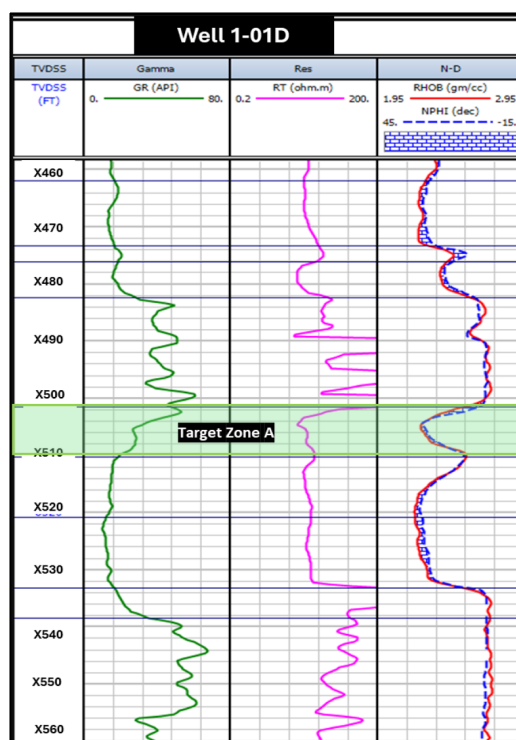


Figure 1—Offset Well 1-01D with triple combo logs identifying the target zone.

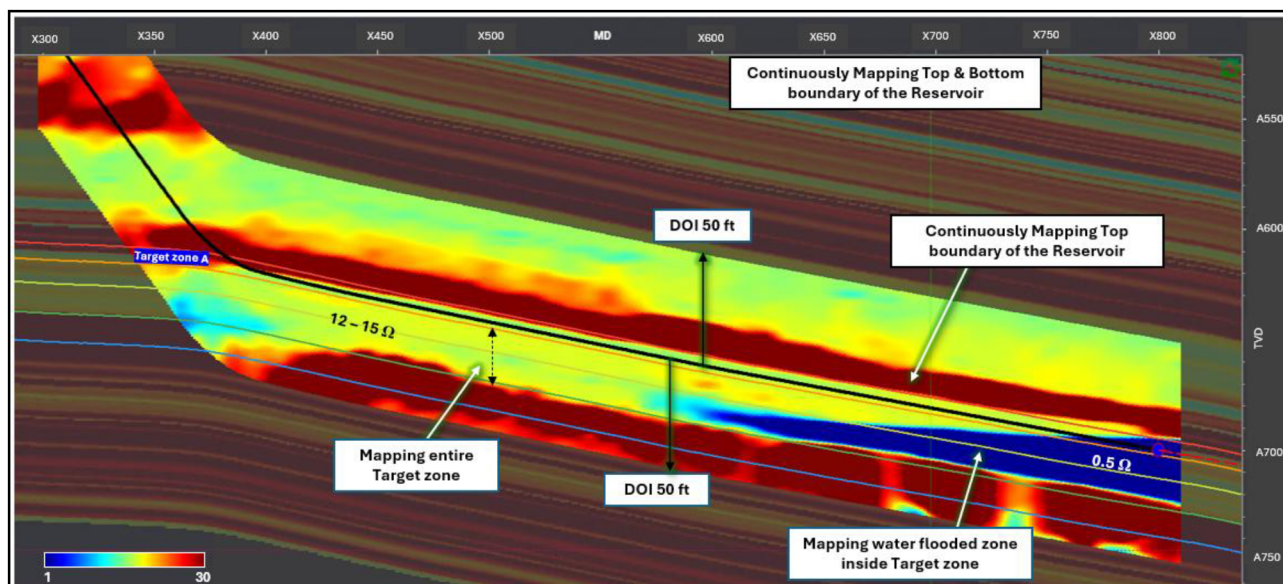


Figure 2—Ultra-deep 1D inversion pre-well modeling Well A, flooded scenario.

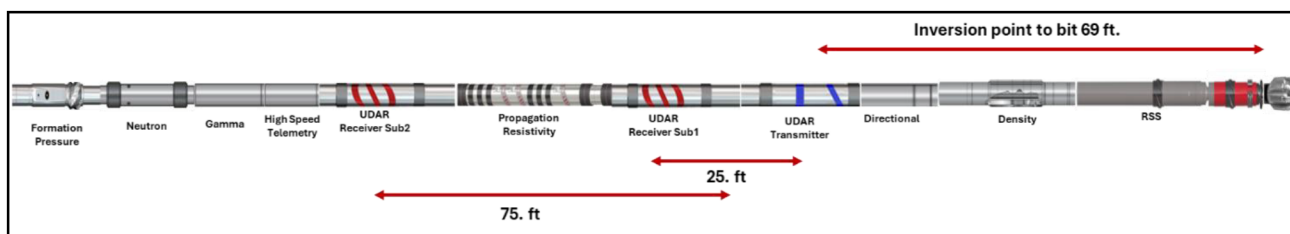


Figure 3—Ultra-deep resistivity schematic bottom hole assembly configuration diagram for 6in section.

Realtime - Drilling Phase

Well A was successfully geosteered and drilled within the optimal target zone. During real-time operations, UltraDeep Resistivity (UDAR) measurements were acquired using both shallow (25 ft depth of investigation) and deep (70 ft DOI) inversion. Upon landing, the UDAR inversion clearly mapped a high-resistivity layer above the wellbore, interpreted as a dense carbonate formation, while the target zone exhibited a homogeneous resistivity profile indicating a consistent and hydrocarbon-bearing interval.

As drilling progressed, both shallow and deep inversions effectively tracked a gradual decrease in resistivity below the wellbore, indicating the approach of a water-bearing zone. There was no possibility to steer upward due to the presence of the dense upper layer an early total depth (TD) was called to avoid intersecting the low-resistivity zone.

Overall, the deep inversion successfully delineated up to six distinct resistivity layers surrounding the wellbore, providing valuable insights into the reservoir fluid distribution. This real-time mapping capability was instrumental in maximizing reservoir contact while minimizing water production risk.

Post-well Analysis

Reservoir engineering objectives to maximize hydrocarbon exposure, minimize water production, and supported sustained well performance were achieved by utilizing UDAR inversion.

Throughout the drilling process UDAR consistently mapped the high resistivity dense layer above the wellbore, which was used as a marker to interpret the formation dipping which was down dip varied from 0.5 to 1.2 degrees. In overall the well was placed at ~4 ft. TVD from top boundary and the distance from the low resistivity reaches up to 4 ft. TVD below the wellbore.

UDAR efficiently mapped the water zone below the wellbore supporting to optimize the completion design for the target well and enhancing oil recovery and sustained production performance.

1D inversion results from both shallow and deep Ultra-Deep Resistivity (UDAR) measurements acquired in real time were compared with memory data, confirming the accuracy and reliability of the real-time interpretation. The analysis revealed that a low- resistivity zone, indicative of water, was approaching the wellbore from the low side.

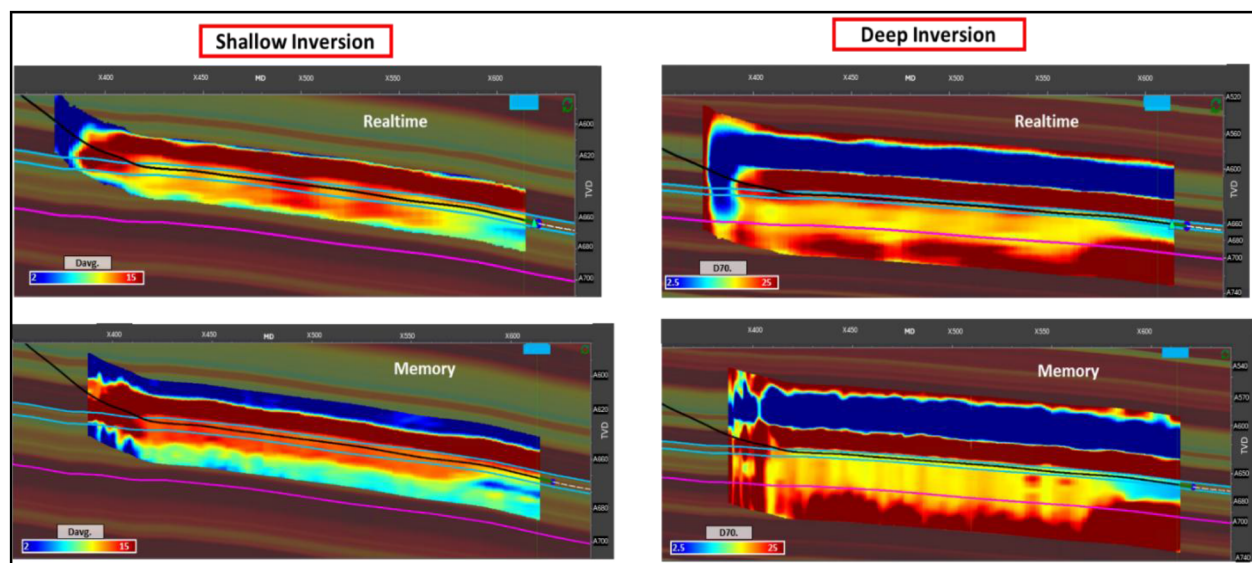


Figure 4—Ultra-deep resistivity comparison Realtime - MEM resistivity 1D inversion.

Well A Production Analysis. The PDA (Post drill Analysis) for Well A confirms a robust post-barge performance in its reservoir, with production, pressure, and PLT data validating the well's optimal placement and reservoir integrity. The well was geosteered using UDAR inversion, achieving 100% reservoir contact, and successfully avoiding conductive zones that could accelerate water encroachment. The well was planned to be drilled as a horizontal single oil producer with a 4,000 ft drain while was completed with 2000 ft drain, due to the water risk mapped by the UDR and completed using LEL .This strategic placement allowed the well to begin producing dry oil up to 4800 BOPD during post barge operations. Based on reservoir sustainability assessments and the need to mitigate long-term risks, the well is currently producing 25003500 BOPD of dry oil for the past ~ 1.5 years. Reservoir pressure in the vicinity was measured between 3,100-3,200 psi, approximately 400 psi above Psat, indicating strong support from nearby injectors and the planned roll-up injectors.

The PLT operation was a critical component of the post-rig evaluation. The main log in the survey was recorded across the interval 13600ft - 16,050 ft, covering the full LEL interval. The survey confirmed uniform oil contribution across the completed interval, with no water cut (WCT) and no gas-oil ratio (GOR) in reservoir condition, indicating clean and efficient production. Notably, CT (Coil Tubing) accessibility reached 100%, surpassing the pre-drill expectation of 92%, which reflects excellent wellbore integrity and completion execution. The PLT also validated the effectiveness of the LEL completion, showing consistent inflow performance and no signs of crossflow or compartmentalization issues. The results were supported by saturation logging and UDAR mapping, which identified a conductive water zone below the drain, prompting an early TD call to preserve well life and production sustainability.

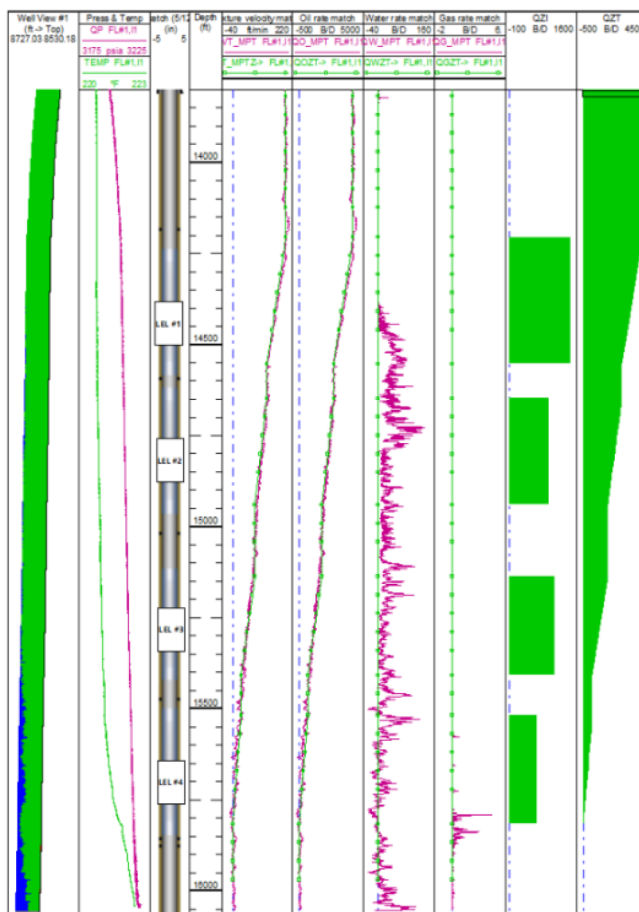


Figure 5—Analysis Flow Profile-flowing Survey, 64/64' choke size.

Future surveillance plans include MRFT prioritization and continued PLT monitoring to track water flood movement and ensure compliance with reservoir pressure guidelines. The well's performance has positioned it as a high-value producer in the ZoneA, contributing significantly to the field's production targets while maintaining reservoir health.

Case Study - Well B

Pre-Well Modeling

The objective was to drill the 6-inch lateral section within Target B, a carbonate layer and map the potential water zone approaching to the wellbore from low side, utilizing Ultra-Deep Azimuthal Resistivity (UDAR), correspondingly would enhance the geosteering to ensure optimal wellbore placement within the most productive zone, away from low-resistivity water-bearing intervals.

Three nearby offset wells were provided Well 01-02D, well 02-02D and Well 03 -02D were used for the pre well modeling stage. Offset Well 01-02D already displays low resistivity inside the target zone and offset Well 02-02D displays a gradual decrease in resistivity towards the bottom of the target zone which was a good reference for the UDAR modeling.

Pre-well modeling confirmed the feasibility of detecting low-resistivity zones beneath the wellbore along the planned trajectory. Several inversion modellings were conducted to determine the most effective BHA configuration.

Therefore, the BHA for the 6-inch included Logging While Drilling (LWD) tools such as UDAR, propagation resistivity and neutron porosity sensors

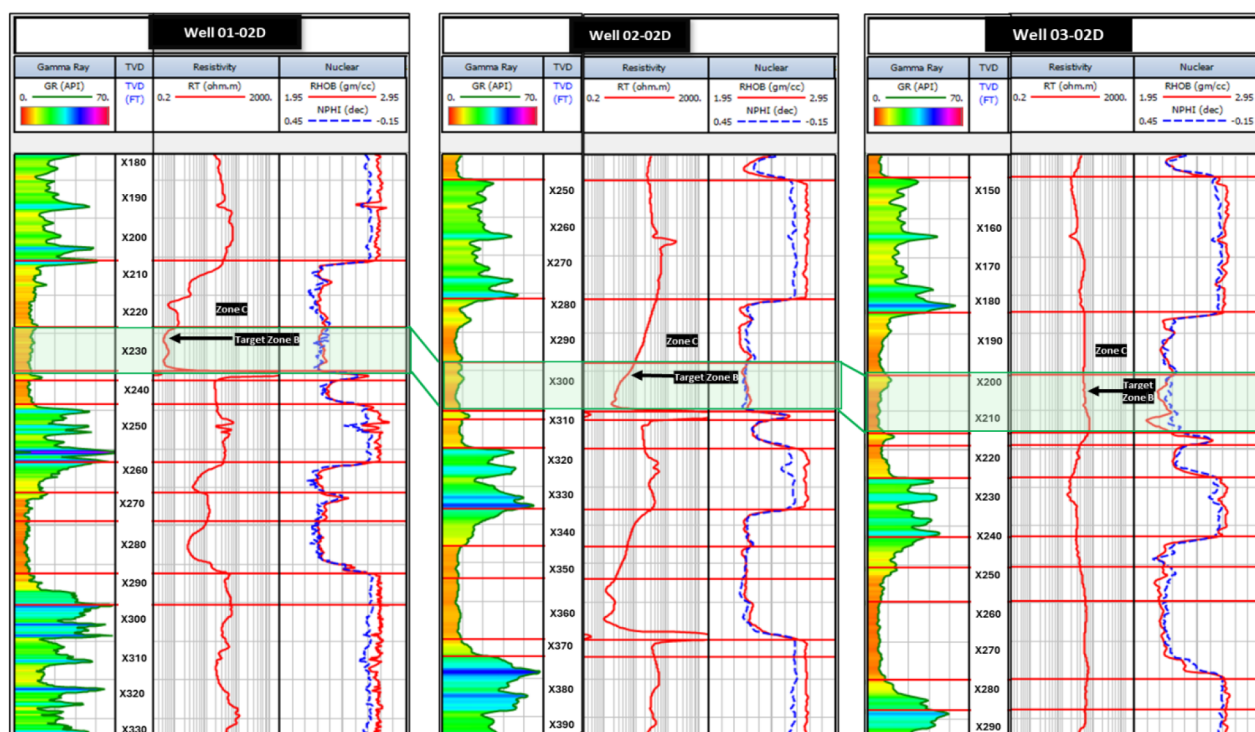


Figure 6—Offset logs Wells 01-02D, 02-02D and 03-02D with triple combo logs.

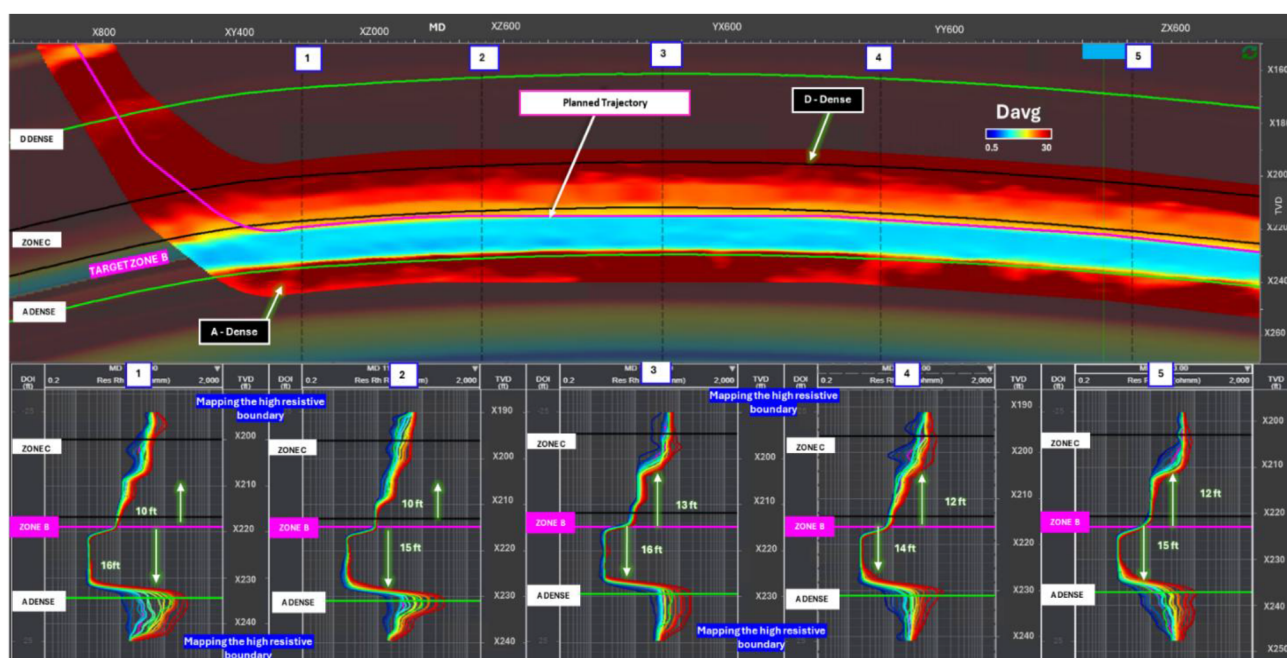


Figure 7—Ultra-deep 1D Inversion resistivity Pre-Well model.

Realtime - Drilling Phase

Well B was successfully geosteered and drilled within the optimal target zone during the 6- inch lateral section. Throughout real-time operations, Ultra-Deep Resistivity (UDAR) 1D shallow inversion served as the primary reference for detecting changes in the resistivity profile around the wellbore.

Upon landing in Target Zone B, the 1D UDAR inversion promptly detected a drop in resistivity below the wellbore, indicating the presence of a water-bearing interval. Simultaneously, the inversion was mapping higher resistivity values within the overlying Zone C, suggesting it was oil-bearing. In response to these

findings, the well trajectory was steered upward into Reservoir Zone C to avoid entering the water zone. Subsequent readings confirmed that Zone B had become flooded, as evidenced by persistently mapped low inverted resistivity values.

As drilling continued within Zone C, shallow inversion data revealed a lateral shift in fluid distribution, with oil saturation reappearing inside Zone B. However, due to ongoing fluctuations in water saturation, the decision was made to maintain the trajectory within Zone C until reaching total depth (TD).

The use of UDAR in real time was instrumental in identifying fluid movement and saturation changes, enabling informed geosteering decisions that helped preserve reservoir quality and reduce water production risk.

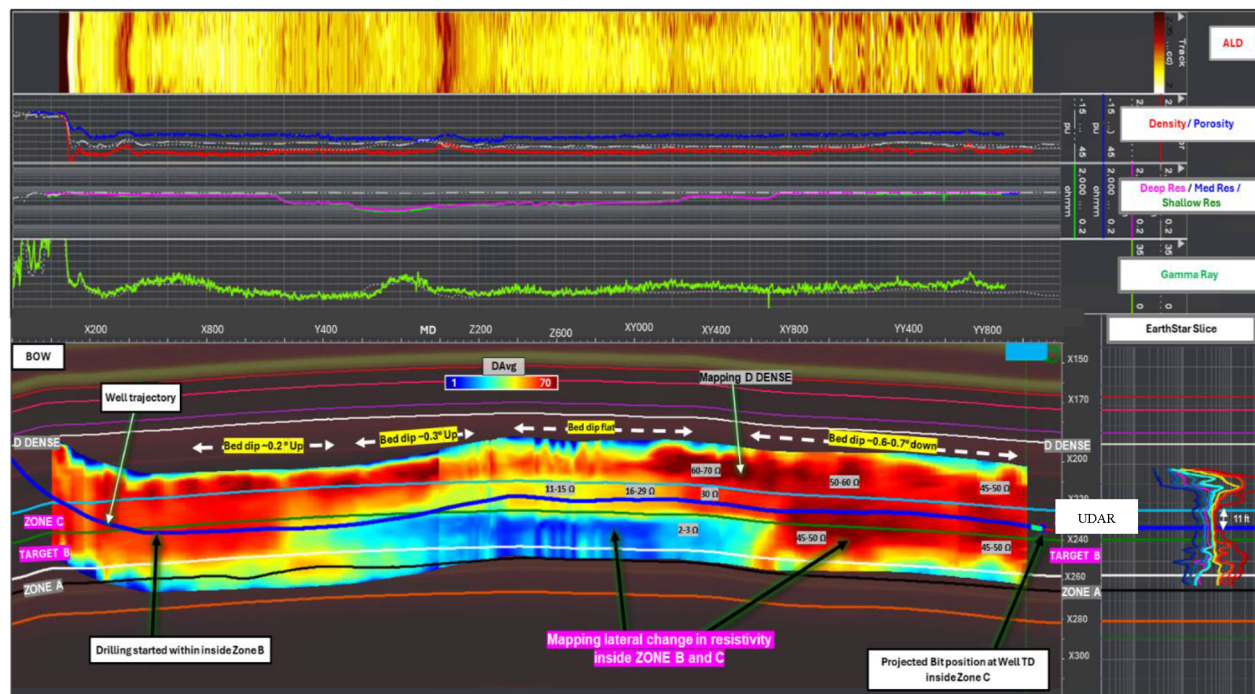


Figure 8—Ultra-deep resistivity inversion real-time main canvas.

Post-Well Analysis

As per Reservoir engineer's anticipation 1D UDAR inversion in Well B, enabled real-time identification of a waterbearing interval preciously in Zone B to adjust trajectory into oil-bearing Zone C, preserving reservoir quality and minimizing water production risk by mapping fluid distribution and guiding completion design to isolate water zones by ICD. The well is now producing from a stable oil interval, contributing to long-term reservoir sustainability.

The 1D Ultra-Deep Resistivity (UDAR) inversion effectively mapped the lateral movement of water around and below the wellbore. It provided a high-confidence interpretation of resistivities inside Target Zone C, which supported the decision to steer the well trajectory upward. Once the well was positioned within Zone C, the inverted resistivity readings were validated by conventional resistivity logging tools, confirming the presence of hydrocarbons.

A comparison between real-time and memory inversion data demonstrated strong consistency in mapping the low-resistivity zone approaching from the lower side of the reservoir. The 1D inversion was able to resolve up to three distinct resistivity layers around the wellbore, offering valuable insight into the reservoir's fluid distribution.

The possibility to map the far field reservoir above and below the wellbore played a key role in proactively decision on realtime and optimizing the completion design, enabling the isolation of water-bearing intervals

and maximizing oil recovery. As a result, the well was completed with the objective of sustaining production and longterm reservoir performance.

Well B Production Analysis. The PDA for Well B confirms a solid post-barge performance in its reservoir, with production, pressure, and PLT data validating the well's strategic placement and reservoir behavior. The well was drilled as a horizontal single oil producer with a 4,600 ft drain and completed using ICD technology to manage inflow and mitigate gas and water risks. Initial production reached up to 5,000 BOPD of dry oil, with a GOR of approximately 1,100 scf/stb and 0% BS&W, confirming clean production conditions. Based on reservoir sustainability assessments and the need to mitigate long-term risks of high-GOR risk zone and proximity to legacy injectors while ensuring long-term reservoir sustainability, the well is currently producing 2500-3500 BOPD of dry oil for the past ~ 1.5 years. Reservoir pressure in the vicinity of Well B was measured between 3,480-3,550 psi indicating good pressure support in the area.

The PLT operation was a critical component of the post-rig evaluation. The tool was deployed via coil tubing and reached a depth of ~12,790 ft MD, covering approximately 50% of the horizontal section. Operational challenges included coil lock-up and non-responsive MAPS sensors due to debris, which limited full drain coverage. Despite these constraints, the PLT confirmed oil production from all logged segments and identified the topmost ICD joint as the primary contributor to surface gas production. Minor crossflow of gas was observed during shut-in, suggesting the well was still stabilizing. These findings were consistent with the presence of free gas in the uppermost part of the well, as also indicated by OHL results.

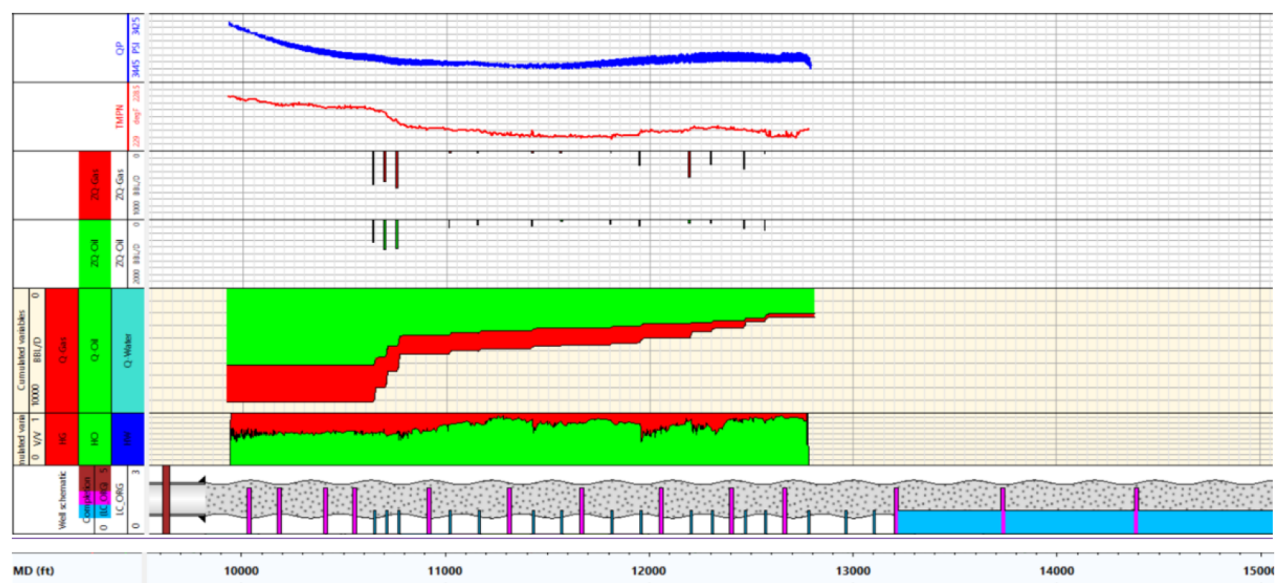


Figure 9—Analysis Flow Profile-flowing Survey, 64/64' choke size.

Future surveillance plans include rigless intervention to mechanically shut off the top ICD intervals contributing to gas production, MRFT prioritization, and continued PLT monitoring to track gas behavior and ensure compliance with reservoir pressure guidelines. The well's performance has positioned it as a valuable producer in the 1B (Zone B), contributing to production targets while maintaining reservoir integrity.

Case Study - Well C

Pre-Well Modeling

The plan was to drill the 6-inch lateral section targeting two reservoirs First Target zone X and second Target Zone Y, utilizing Ultra-Deep Azimuthal Resistivity (UDAR), to map lateral resistivity variations

within Zone Y while drilling through Zone X. This approach aimed to assess the feasibility of geosteering into Zone Y if favorable conditions were identified, while also ensuring precise well placement within oil-bearing intervals overcoming structural and geological complexities.

Two main offset logs were utilized during the modeling stage Well 01-01D and Well 02-01D, analyzing their resistivity properties inside both target zones. Well 01-01D was identified as representative of the anticipated reservoir resistivity profile due to the presence of low resistivity zone inside the target zone Y and was used for reference and correlation with the real-time data during drilling.

Extensive modeling inversion was conducted to determine the inversion response for the possible scenario, The modeling confirmed UDAR's capability to accurately detect lateral resistivity changes both above and below the wellbore, providing critical support for real-time decision-making. The most effective UDAR settings (optimal transmitter-receiver spacing and operating frequency) and BHA configuration were determined in the pre modeling stage.

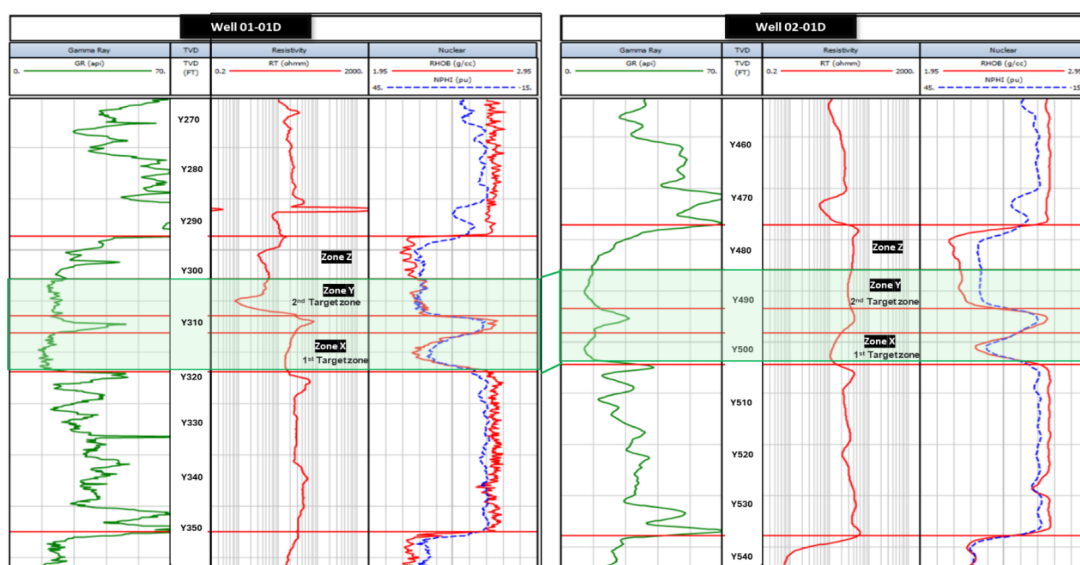


Figure 10—Offset logs Wells 01-01D and 02-01D with triple combo logs.

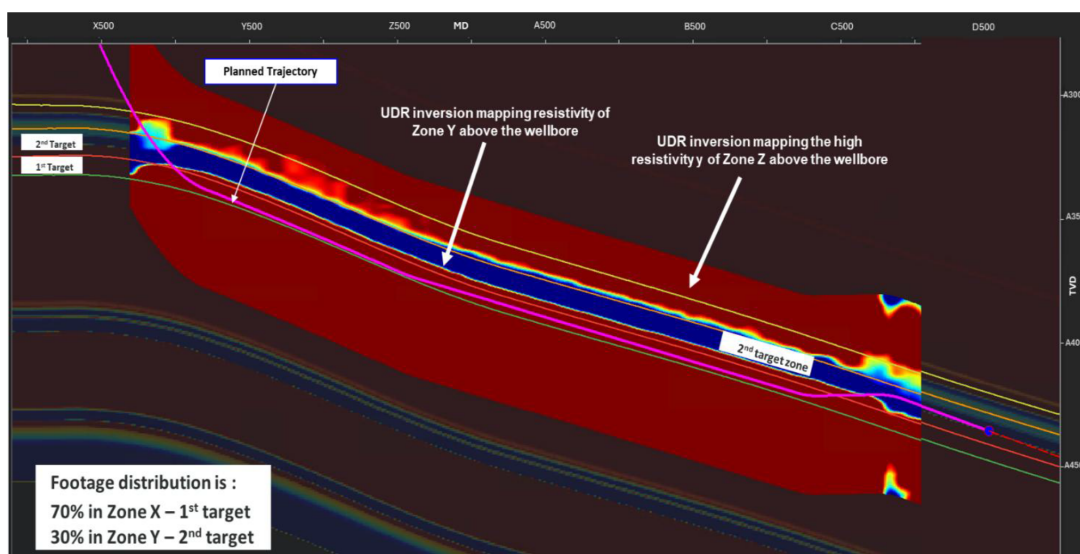


Figure 11—Ultra-deep resistivity inversion pre - model, low resistivity zone inside target zone Y.

Realtime - Drilling Phase

Well C was effectively drilling and steered achieving well objectives to drill in Target zone X and Y. Throughout the lateral section UDAR 50ft. depth of investigation served as the primary inversion to detect and identify the fluid distribution around the wellbore.

The well was landed the first target zone X, the well path intersected reservoir Y, where conventional resistivity tools confirmed the presence of a low-resistivity interval, indicating potential fluid contact.

Once the well landed inside Zone X, UDAR mapped the presence of water bearing zone within the second target zone Y, located above the wellbore. As drilling progressed through approximately 25% of the planned lateral section, UDAR inversion results revealed a lateral transition from water-bearing to oil-bearing zones within Zone Y.

Continuous real-time UDAR monitoring above the wellbore confirmed the presence of hydrocarbons. Based on this data, the well trajectory was adjusted upward to access the oil-bearing portion of Zone Y. The final 30% of the lateral section was drilled within this zone, aligning with the well objectives. Conventional resistivity tools validated the UDAR inversion results, confirming the presence of oil in Zone Y.

The integration of UDAR in real-time was critical for dynamic decision-making, enabling the identification of fluid boundaries and optimizing reservoir contact. This approach allowed for effective production planning from both target zones, maximizing hydrocarbon recovery.

Post-well Analysis

As per Reservoir Engineer's well placement guidelines Well C was placed successfully within oil-bearing intervals of Zones X and Y while avoiding low-resistivity, water-prone layers. The well is strategically positioned across the most productive sections of both zones and is contributing significantly to field production.

The 1D Ultra-Deep Resistivity (UDAR) tool played a pivotal role in optimizing well placement by accurately identifying oil-bearing zones and steering the well trajectory away from low-resistivity (water-bearing) intervals (Fig 12). UDAR inversion efficiently mapped the lateral change in resistivity inside Zone Y from water-bearing to oilbearing zones. This high-confidence interpretation enabled a critical decision to steer the wellbore upward, aligning with reservoir objectives. Once the well was positioned within Zone Y, conventional resistivity logging tools validated the inverted resistivity readings, confirming the presence of hydrocarbons.

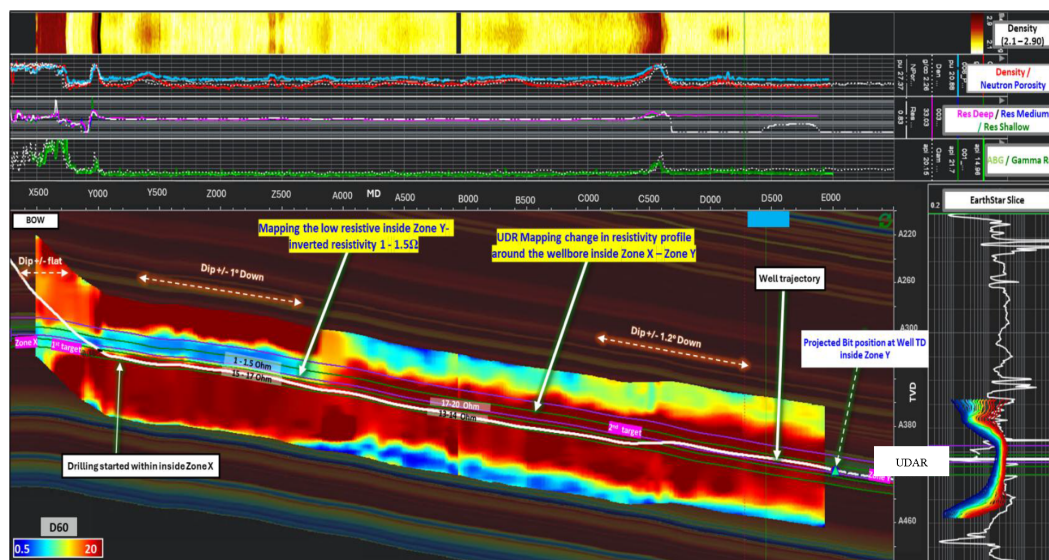


Figure 12—Ultra-deep resistivity inversion real-time main canvas.

A comparison between real-time and memory inversion demonstrated strong consistency in mapping resistivity changes, reinforcing the reliability of UDAR data. The 1D inversion was able to resolve up to three distinct resistivity layers around the wellbore, offering valuable insight into the reservoir's fluid distribution.

In addition, 3D UDAR inversion provided (Fig.13) a 360-degree view around the wellbore. This top-down perspective revealed a gradual increase in resistivity across Zone Y, potentially associated with lateral strike-slip movements in the formation. This spatial understanding enhanced the geological interpretation and supported the strategic placement of the well within the most productive intervals.

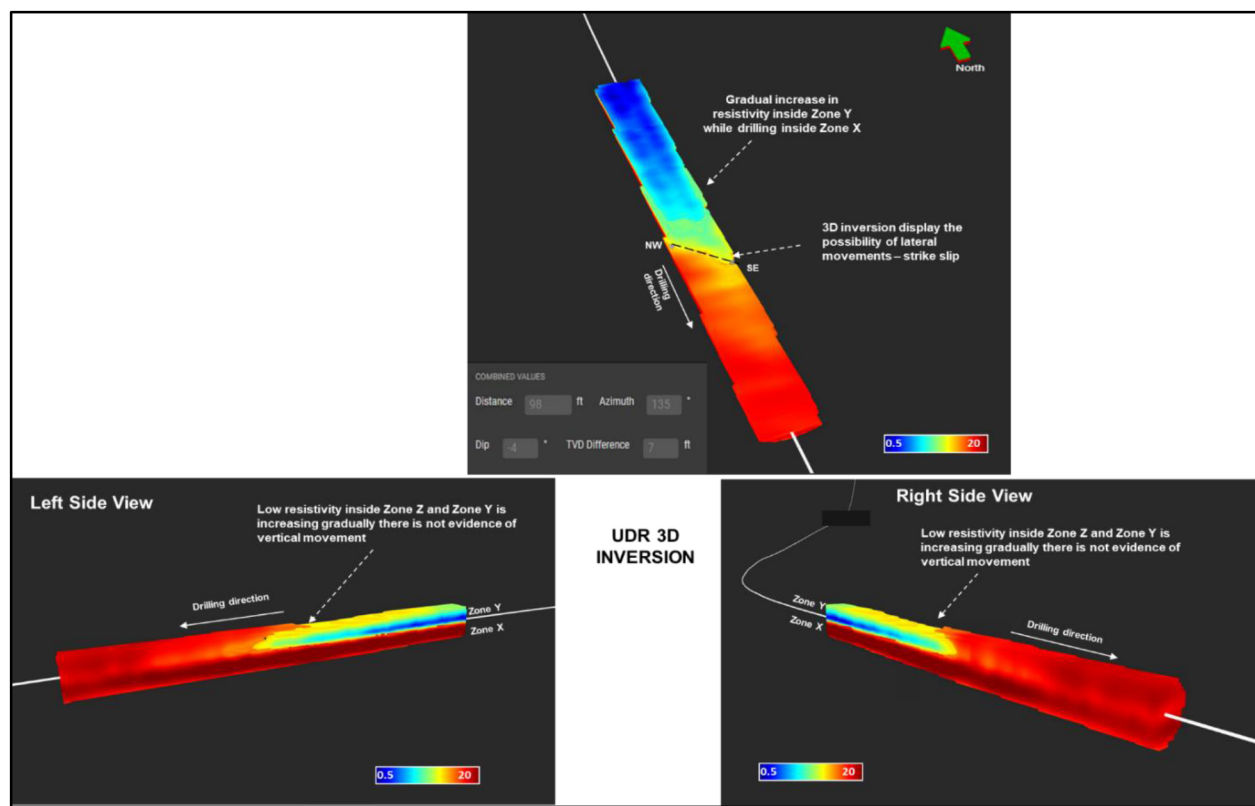


Figure 13—Ultra-deep 3D Inversion - Top Down, left and right-side view.

Well C Production Analysis. The production analysis for well C confirms a solid post-barge performance in its reservoir, with production and pressure data validating the well's strategic placement and reservoir behavior. The well was drilled as a 5800 ft horizontal oil producer targeting dual intervals—X and Y—and completed using PPL/AICD technology to manage inflow and mitigate gas and water risks. Initial production aligned with expectations, with dry oil up to 2500 BOPD and a GOR of approximately 2,500 scf/stb during the past 8 months, confirming stable reservoir conditions as per the expected well location. Reservoir pressure in the vicinity was measured at approximately 3,500 psi, indicating strong support from nearby injectors and planned roll-up injectors. PLT operation was not conducted due to accessibility issues.

The MDT operation was a critical component of the well evaluation and proper well completion. In layer X, the reservoir exhibited an average porosity of 21%, water saturation of 21%, pressure of 3,493 psi, and mobility of 3.2 md/cp, indicating medium oil. In layer Y, porosity averaged 23%, water saturation was 11%, pressure was 3,457 psi, and mobility reached 6.8 md/cp, indicating light oil. These results confirmed consistent inflow performance with no signs of crossflow or compartmentalization. No major anomalies were observed, and the well demonstrated moderate to good reservoir properties across both targets.

Future surveillance plans include MRFT prioritization to track reservoir behavior, manage drawdown, and ensure compliance with pressure management guidelines. The well's performance has positioned it as a valuable contributor in its reservoir, supporting production targets while maintaining reservoir integrity and long-term sustainability.

Summary & Conclusion

The application of Ultra-Deep Azimuthal Resistivity (UDAR) technology in the three case studies presented from the offshore carbonate field has proven to be a critical enabler in transforming reservoir mapping, geosteering precision and lower completion optimization. Through a combination of advanced pre-well modeling, real-time Resistivity inversions and Post-Well validation, the use of UDAR tools led to enhanced hydrocarbon recovery, avoidance of premature water breakthrough and considerable cost efficiencies. Collectively, these outcomes contributed to an estimated value creation of USD 294 million in 2024 alone—an outcome that strongly validates the business case for deploying such technology in mature and complex reservoir environments.

Across all three wells (A, B and C), UDAR was instrumental in proactively identifying low-resistivity, waterbearing intervals before they were intersected, enabling timely course corrections in the well trajectory. This capability not only optimized reservoir exposure but also informed intelligent blanking strategies during completion design, thereby preserving the long-term integrity and productivity of the wells. The technology's ability to deliver actionable subsurface insights at distances up to 100 ft from the borehole provided a competitive advantage over conventional near-bit measurements, particularly in structurally complex carbonate reservoirs where thin beds, lateral fluid variations, and ambiguous resistivity responses are the norm.

- In Well A, the UDAR tool effectively mapped a conductive zone trending upward from beneath the wellbore, prompting an early Total Depth (TD) call. This decision avoided intersection with a water-bearing interval, preserving oil production and extending the well's economic life. Importantly, the 1D inversion was able to resolve multiple resistivity layers surrounding the wellbore in real time, offering granular fluid distribution data that was later validated by memory data sets. The upper dense carbonate zone, consistently mapped throughout the well, served as a key structural marker that anchored the Geosteering decision-making process.
- Well A is a top performer, producing 2,500-3,500 BOPD of dry oil for 1.5 years after peaking at 4,800 BOPD. UDAR geosteering ensured full reservoir contact and avoided water zones. Completed with 2,000 ft LEL due to water risk. PLT confirmed clean, uniform oil flow and excellent well integrity. Reservoir pressure is strong (3,100-3,200 psi), supported by injectors. Ongoing surveillance includes MRFT and PLT to sustain performance.
- In Well B, UDAR's real-time data revealed an active water-front below the planned trajectory while concurrently detecting an oil-bearing interval above. The ability to clearly differentiate between these layers enabled the team to make an informed decision to steer upward into a secondary target zone, which was ultimately confirmed to be hydrocarbon-bearing. The dynamic water movement captured during real-time operations would not have been detected using shallow resistivity measurements alone. This case highlighted the value of UDAR not only as a resistivity mapping tool but also as a strategic asset for mitigating reservoir conformance issues and improving completion efficiency.
- Well B has shown strong performance since drilling, producing clean oil with minimal gas and water issues thanks to its horizontal design and ICD completion. Initial output peaked at 5,000 BOPD, and it has consistently delivered 2,500-3,500 BOPD over the past 1.5 years, supported by healthy reservoir pressure (3,480-3,550 psi).

- Well C presented a more complex structural scenario, where the trajectory was designed to intersect two distinct target zones, with a known conductive interval above the initial trajectory. Here, UDAR's strength in lateral mapping came to the forefront. As the well progressed through the first reservoir, Zone X, the tool detected a gradual lateral increase in resistivity in Zone Y, indicating a transition from water- to oil-bearing rock. This key observation led to a deliberate trajectory adjustment upward into the more productive portion of Zone Y. Notably, the 3D UDAR inversion provided a 360° spatial resistivity profile, revealing subtle resistivity anisotropies and suggesting potential strike-slip movements in the formation—insights that significantly enriched the geological interpretation. The integration of both 1D and 3D UDAR data empowered a level of decision-making precision that has not been achievable with traditional geosteering workflows.
- Well C has delivered strong post-drill results, confirming strategic placement and stable reservoir behavior. The 5,800 ft horizontal producer, completed with PPL/AICD, targets two oil-bearing intervals. It has consistently produced ~2,500 BOPD with a GOR of ~2,500 scf/stb with no water, supported by nearby injectors maintaining ~3,500 psi pressure. Zone X shows medium oil (21% porosity, 21% Sw, 3.2 md/cp), while Zone Y holds light oil (23% porosity, 11% Sw, 6.8 md/cp).

In all three wells, the deployment of UDAR tools led to confident decision-making regarding well trajectory and completion design—decisions that were directly linked to enhanced productivity and reduced water cut, which led to substantial value creation in production, amounting to more than \$150 million in 2024. The post-rig production performance of the wells, all of which flowed dry oil without the need for artificial lift, further underscores the effectiveness of the technology. This is particularly striking when compared to nearby offset wells that suffered from water encroachment, flow cessation or ineffective completions due to the absence of advanced reservoir mapping solutions.

From an operational standpoint, the successful implementation of UDAR also streamlined post-rig activities. The clarity provided by the inversion results enabled more accurate blanking of water-prone intervals, reducing the risk of completion failure and subsequent interventions. In the context of mature field development, where well intervention costs and productivity risks are significantly higher, this optimization has profound implications for long-term field economics.

Furthermore, the integration of UDAR into geosteering workflows fostered stronger cross-functional collaboration between drilling, reservoir, and completion teams. The shared interpretation of high-resolution inversion data in real time allowed for a unified operational approach, minimizing ambiguity and fostering proactive responses to subsurface changes. These collaborative dynamics not only accelerated the decision-making process but also improved operational agility in the face of geological surprises.

Beyond the technical achievements, this study demonstrates a clear shift in the operational paradigm for mature carbonate field development. Rather than relying solely on reactive decisions from shallow measurements, the industry now has access to tools that enable proactive steering based on deeper and broader resistivity mapping. This shift enhances the probability of meeting well objectives, especially in reservoir settings with high heterogeneity, water conformance challenges, and minimal stratigraphic markers.

Another significant takeaway from this study is the reliability and repeatability of UDAR's inversion outputs. In all three cases, a strong correlation was observed between real-time and memory resistivity data, affirming the consistency of the tool's performance. The inversion modeling conducted in the pre-well phase was also validated during operations, indicating the robustness of the simulation methodologies used for tool configuration and reservoir response prediction. These validations lend strong confidence to future UDAR-based workflows and support their broader deployment across similar carbonate fields.

In summary, the implementation of Ultra-Deep Azimuthal Resistivity tools across these three wells in the offshore field has established a new benchmark for reservoir mapping, geosteering effectiveness and well completion optimization. The demonstrable production gains and well life extension achieved through

real-time, deep resistivity inversion not only validate the technology but also emphasize the importance of integrating such advanced measurements in future field development strategies. As the industry continues to face tighter economic margins, declining reservoir productivity and increasingly complex geological settings, the ability to reduce uncertainty, optimize contact and avoid water production will become even more critical. UDAR has shown that it is not merely a logging tool but a transformative enabler of smarter, more profitable field development.

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